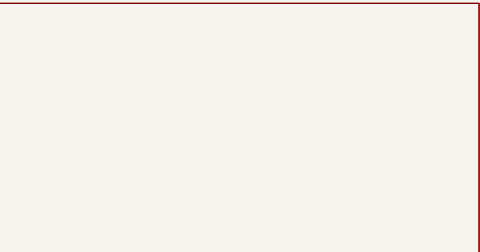


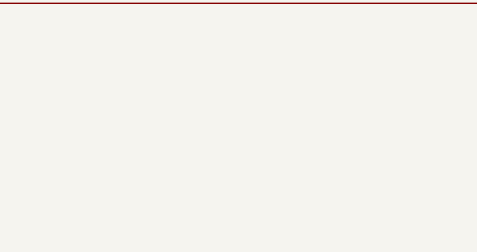
TestBoard1 : VAR-SOM-MX8M-PLUS proof-of-concept carrier

SoM Connector (VAR-SOM-MX8M-PLUS)



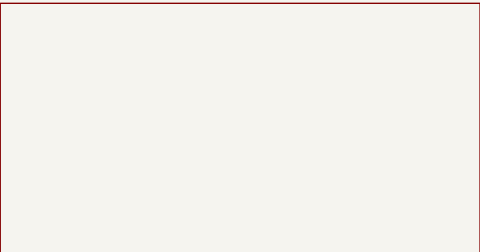
File: SOM.kicad_sch

Power input / regulators / status LEDs



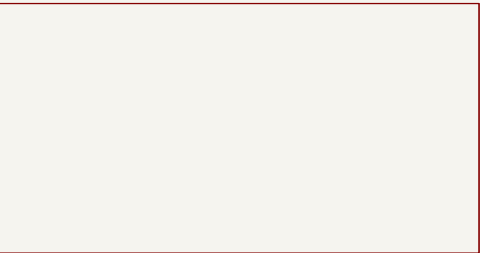
File: Power.kicad_sch

Gigabit Ethernet (KSZ9031 RGMII PHY + MagJack)



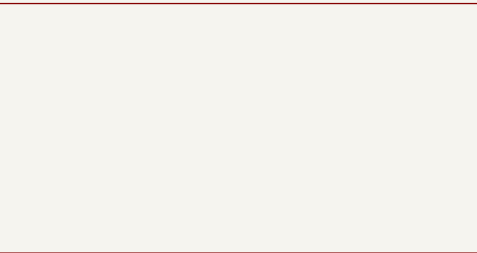
File: Ethernet.kicad_sch

USB 2.0 host ports (front + back)



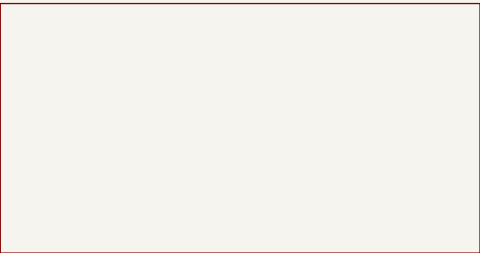
File: USB.kicad_sch

I2C / SPI / microSD / RGB LEDs / debug UART



File: Peripherals.kicad_sch

Raspberry Pi 40-pin HAT header (J9)



File: RPI_Header.kicad_sch

Proof-of-concept carrier for the Flux project. Features per Readme.md:
Ethernet (back), 2x USB2.0-A host (front+back), I2C & 2x SPI IDC, microSD (front),
RPi 40-pin HAT header, 12V input with on-board 5V/3V3 regulators, power status LEDs,
3x addressable RGB LEDs. 4-layer PCB, whole-mm outline, 4x M3 mounting holes 4mm from edges.

OpenTools / Flux			
Sheet: /			
File: TestBoard 1.kicad_sch			
Title: TestBoard1 - VAR-SOM-MX8M-PLUS carrier (root)			
Size: A3	Date:		Rev: A
KiCad E.D.A. 10.0.2	Id: 1/7		

VAR-SOM-MX8M-PLUS 200-pin SO-DIMM. VCC_SOM<-+5V, VDD_ENET<-+3V3, unused display/audio = NC.

U1A
VAR-SOM-MX8M-PLUS

U1B
VAR-SOM-MX8M-PLUS

U1C
VAR-SOM-MX8M-PLUS

U1D
VAR-SOM-MX8M-PLUS

U1E
VAR-SOM-MX8M-PLUS

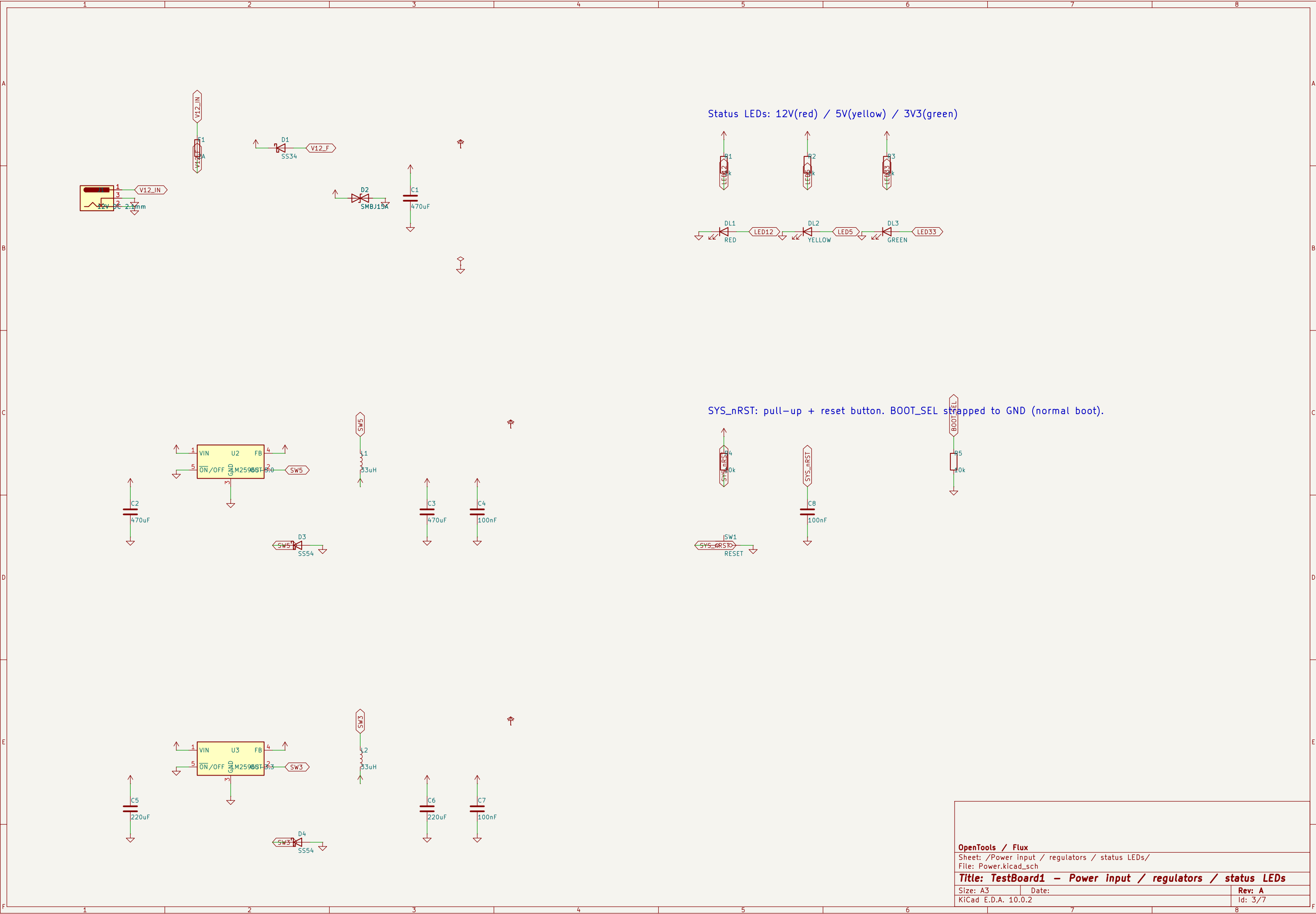
U1F
VAR-SOM-MX8M-PLUS

U1G
VAR-SOM-MX8M-PLUS

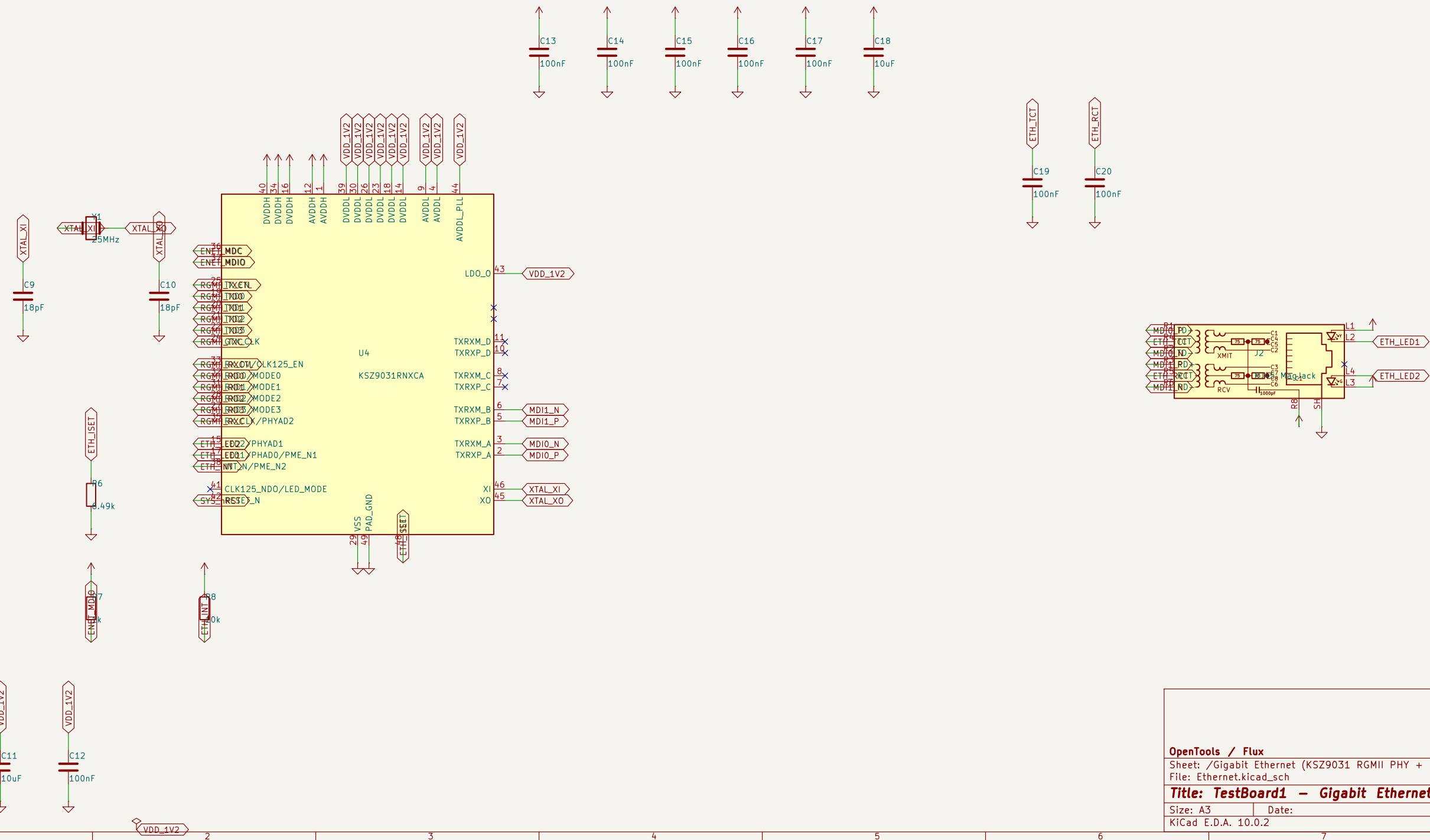
U1H
VAR-SOM-MX8M-PLUS

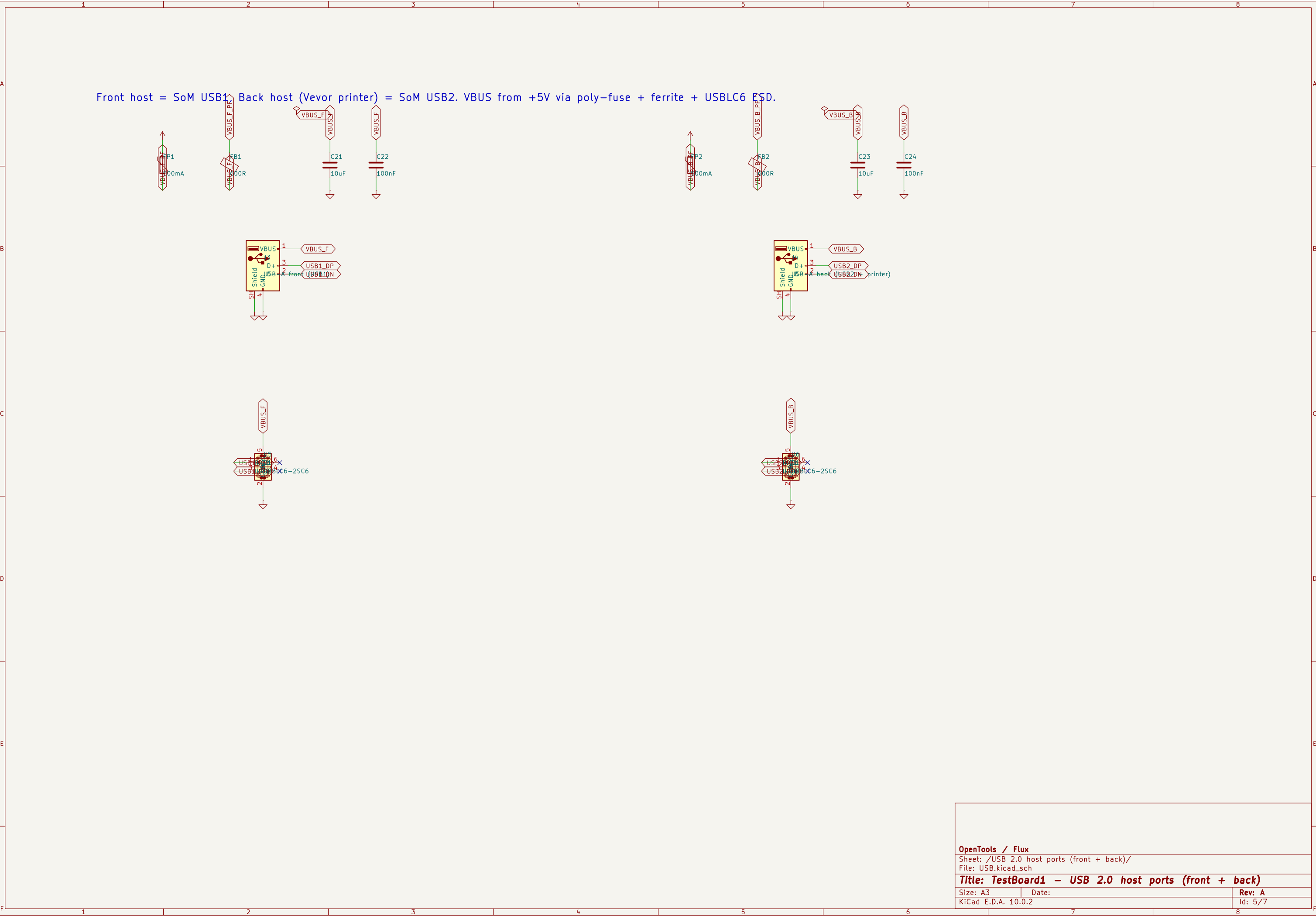
U1I
VAR-SOM-MX8M-PLUS

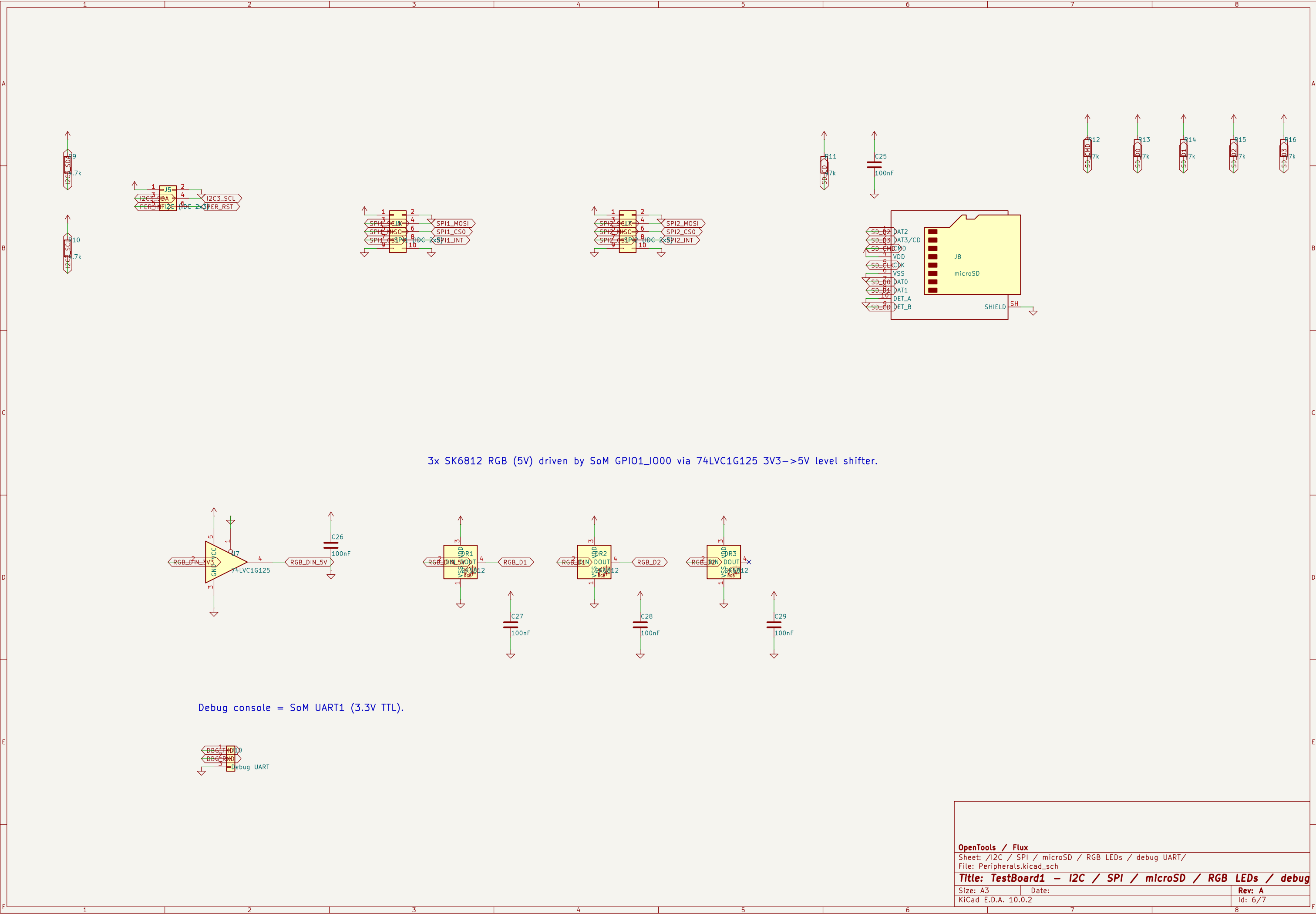
OpenTools / Flux
Sheet: /SoM Connector (VAR-SOM-MX8M-PLUS)/
File: SOM.kicad_sch
Title: TestBoard1 - SoM Connector (VAR-SOM-MX8M-PLUS)
Size: A2
Date:
Rev: A
Id: 2/7



KSZ9031 RGMII PHY. MDI pairs A/B -> MagJack TD/RD (10/100/1000 auto). PHY straps at defaults; verify PHYAD before fab.







3x SK6812 RGB (5V) driven by SoM GPIO1_I000 via 74LVC1G125 3V3→5V level shifter.

Debug console = SoM UART1 (3.3V TTL).

RPi-style 40-pin header. Power(3V3/5V/GND), I2C1->SDA1/SCL1, SPI2 (shared bus w/ J7),

UART3->TXD/RXD, I2C4->HAT-ID EEPROM (pin27/28), + free GPIO. Lines with no free SoM GPIO = NC.

